INDEPENDENT PROJECT ANALYSIS · COMMITTEE BRIEFING

Requirement *Pilot*

An evidence-grounded, human-in-the-loop, agentic requirements engineering platform — deep analysis of the reference document dated 30 September 2026.

CONDITIONAL GO · CONSOLIDATED SCORE 4.0 / 5

RP

Analysis briefing · October 2026

Prepared for the supervisor and defense committee

Companion to the 27-page written analysis report (11 sections · 5 figures · 10 tables)

AGENDA

Briefing roadmap

01 **Verdict at a Glance**
What the platform is, and how it scores across six assessment dimensions.

02 **Novelty & Contribution**
Ten planned contributions; two defensible research cores (N2, N7), eight disciplined increments.

03 **Feasibility, Architecture & Risk**
Sound, conservative design; schedule arithmetic with zero buffer; eleven risks, four newly identified.

04 **Actions & Recommendation**
Twelve actions across three time horizons, the descope ladder, and four conditions attached to the go.

Verdict at a Glance

“An exceptionally well-prepared graduation project — honest about the literature, precise about its mechanisms, disciplined about evaluation.”

6 dimensions scored

4 conditions attached

No structural flaws found

What is RequirementPilot?

Rather than generating polished requirements from a single prompt, RequirementPilot conducts structured stakeholder interviews, distinguishes explain-why questions from extraction tasks, and grounds every requirement in versioned evidence. **Humans approve each transition — no requirement gap is ever silently filled by a model.**

10

planned contributions (N1–N10)

7

research questions with quantitative metrics

8

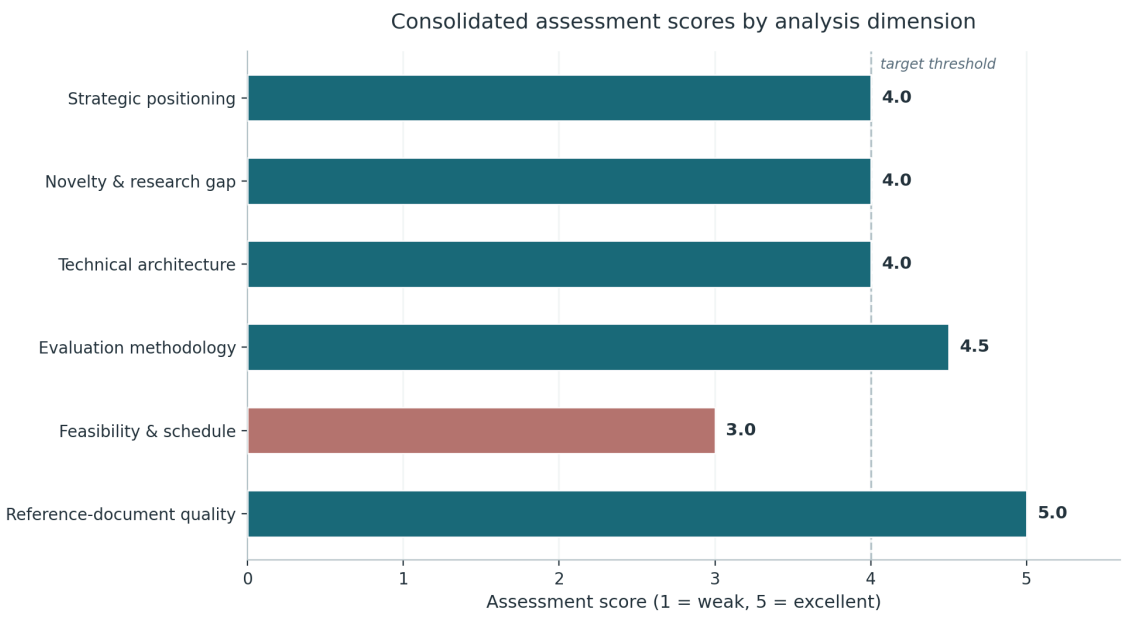
recent systems benchmarked in the literature review

3

evidence languages — English, Egyptian Arabic, code-switched

Team of four · 6-month (24-week) graduation project window · Integration-first positioning that explicitly forbids overclaiming novelty.

Consolidated verdict — conditional go at 4.0 / 5



Strategic positioning — 4.0
integration-first, honestly framed / framing A–D not yet chosen

Novelty & research gap — 4.0
N2 + N7 defensible cores / self-assessed matrix, no search protocol

Technical architecture — 4.0
determinism-first, authority-aware retrieval / context, concurrency, observability unspecified

Evaluation methodology — 4.5
strategies, not models; full ablation map / no power analysis or IAA plan

Feasibility & schedule — 3.0
clear MVP and M1 gate / 26 weeks of work in 24; benchmark last

Reference-document quality — 5.0
comprehensive, honest, actionable / none material

Proceed — subject to four conditions, detailed in Part 04.

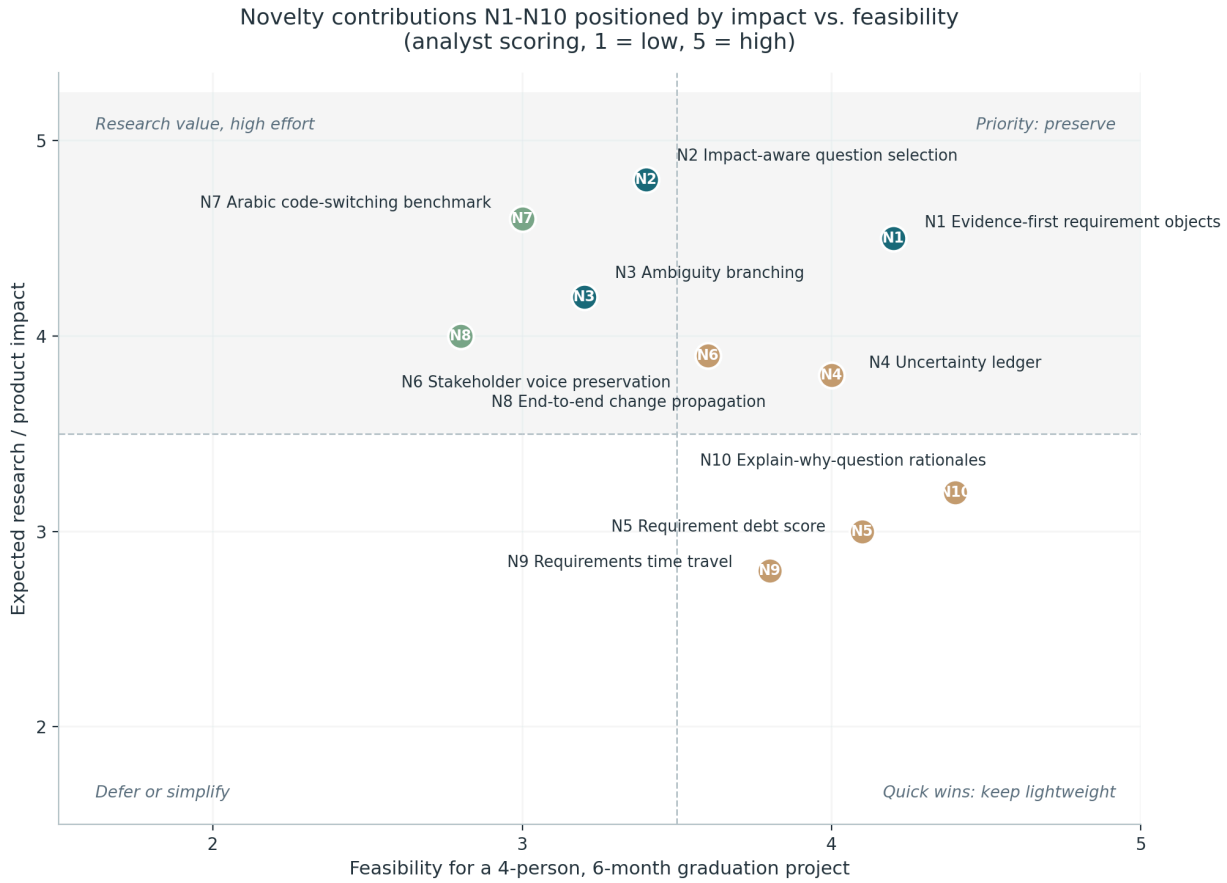
Novelty & Contribution

Two defensible research cores — impact-aware question selection and an Arabic code-switching benchmark — carried by eight useful but individually incremental contributions.

Assessment basis: analyst scoring of the ten contributions against recent systems (RECOVER, LENS, OntoAgent, ReqElicitGym, ProReFiCIA, AraREQ, MAAD, MedNegotiator).

SECTION 02 · NOVELTY & CONTRIBUTION

Ten contributions positioned by impact vs. feasibility



PROTECT — core of the thesis

N2 Impact-aware question selection · **N7** Arabic / code-switched benchmark — individually defensible, ablation-critical, independently publishable.

KEEP — lightweight

N1 Evidence-first objects · **N3** Ambiguity branching · **N4** Uncertainty ledger · **N5** Requirement debt score · **N6** Voice preservation · **N10** Rationale explanations — valuable engineering, not headline research.

DEFER — descope candidates

N8 Cross-artifact change propagation · **N9** Requirements time travel — Phase B material; first cuts when time compresses.

Scores are analyst estimates on a 1–5 scale; the matrix is self-assessed by the team, hence Action 2 (systematic search protocol).

SECTION 02 · NOVELTY & CONTRIBUTION

The defensible core: N2 and N7

N2 — Impact-aware question selection

impact 4.8 | feasibility 3.4

- Deterministic utility scoring decides which question the interviewer asks next — no black-box agent loop
- The system's central claim stands or falls with this mechanism: ablation-critical
- **Weakness:** utility weights are currently unjustified guesses → pilot ablation on the first 20 scenarios in month 3 (Action 7)

N7 — Egyptian Arabic / code-switched benchmark

impact 4.6 | feasibility 3.0

- First requirements benchmark of its kind for Egyptian Arabic and code-switching; independently publishable
- Aligned with team skills — dedicated NLP/Arabic member — and with the local context
- **Weakness:** currently scheduled in months 5–6, too late to evaluate properly → production from month 2 (Action 6)

Why these two are defensible: both are measurable, both have drawn boundaries (exact dialect, task, benchmark limits), and both match available skills and data.

03 / 04

Feasibility, Architecture & Risk

Sound, deliberately conservative engineering — set against schedule arithmetic with zero buffer and a risk cluster concentrated in execution, not in the idea.

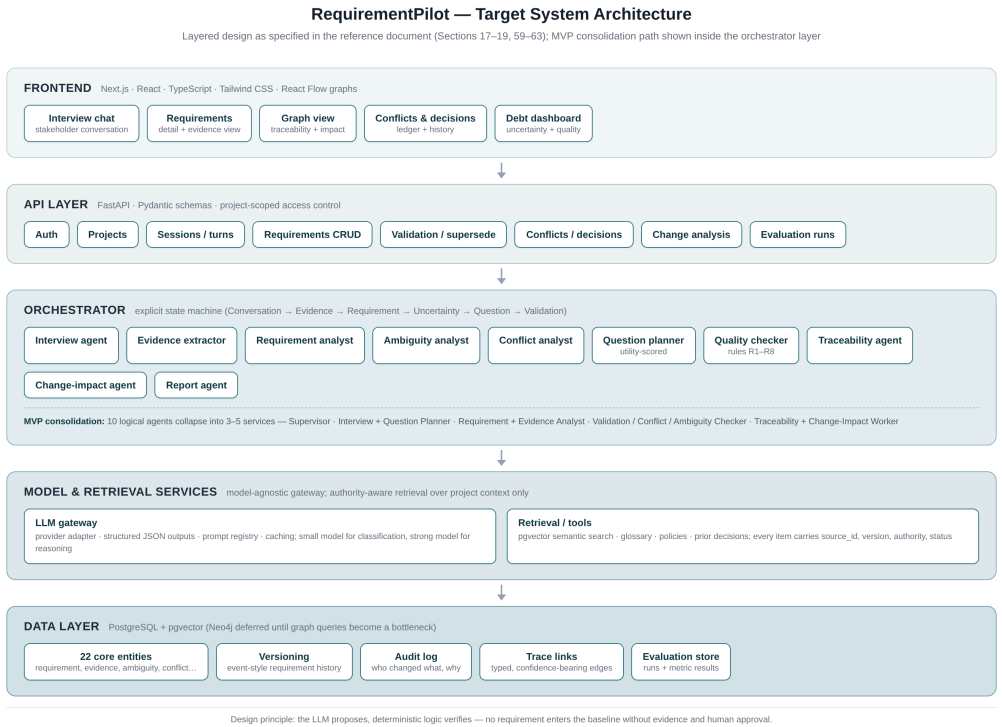
Architecture 4.0 / 5

Schedule 3.0 / 5

11 risks mapped

03

Architecture: conservative by design — and right



What is strong

- ✓ Explicit state machine instead of a heavyweight agent framework — no hidden non-determinism in a system whose thesis is determinism
- ✓ Evidence is a first-class object: **22-entity** model, event-style requirement versioning (never overwrite; author, source, reason, timestamp stored)
- ✓ Authority-aware retrieval: approved policies (**authority 1.00 / 0.95**) cannot be silently overridden by AI inference (**0.3×**)
- ✓ Eight hallucination controls wrapped around the human approval gate.

What is unspecified

- ✕ Context management, concurrency, and observability are not yet designed → run graph-query and infrastructure spikes before Phase 3 (Action 5).

SECTION 03 · FEASIBILITY, ARCHITECTURE & RISK

Evaluation methodology — the strongest dimension (4.5 / 5)

4.5

7 research questions

Each RQ carries named metrics — extraction accuracy (RQ1), utility of asked questions, conflict-detection quality, trace-link precision — and the design evaluates interview strategies, not model impressions.

Disciplined baselines

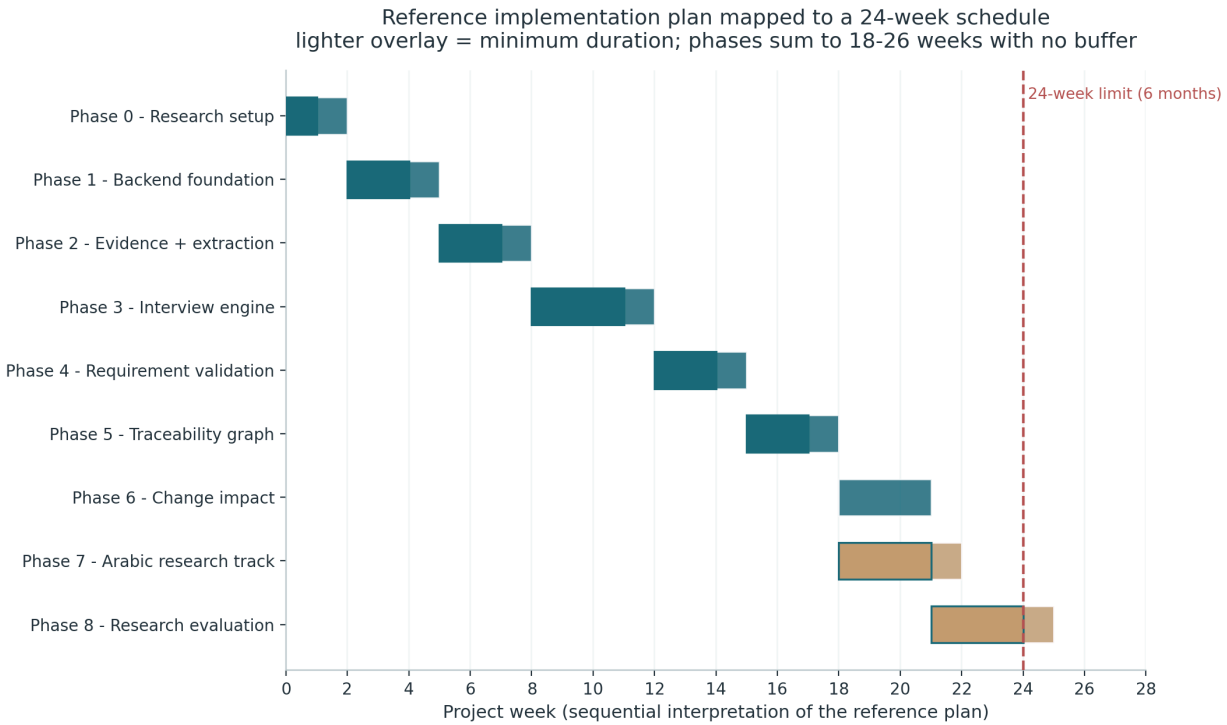
Four interview strategies — single-shot generation, unrestricted conversational LLM, ontology-guided interviewer, impact-aware planner — crossed with a four-language matrix, plus three artifact-validation baselines.

Layered data strategy

Public corpora first: PURE (**79 documents, 34,268 sentences**), PROMISE and SWARD where licensing permits; ReqElicitGym (**101 scenarios**) for baseline runs; real conversations only where permissions allow.

Missing before any data collection: statistical power analysis, random seeds, significance testing plan, and an inter-annotator agreement threshold → fixed by Action 3 (weeks 1–2).

Schedule feasibility — the weakest dimension (3.0 / 5)



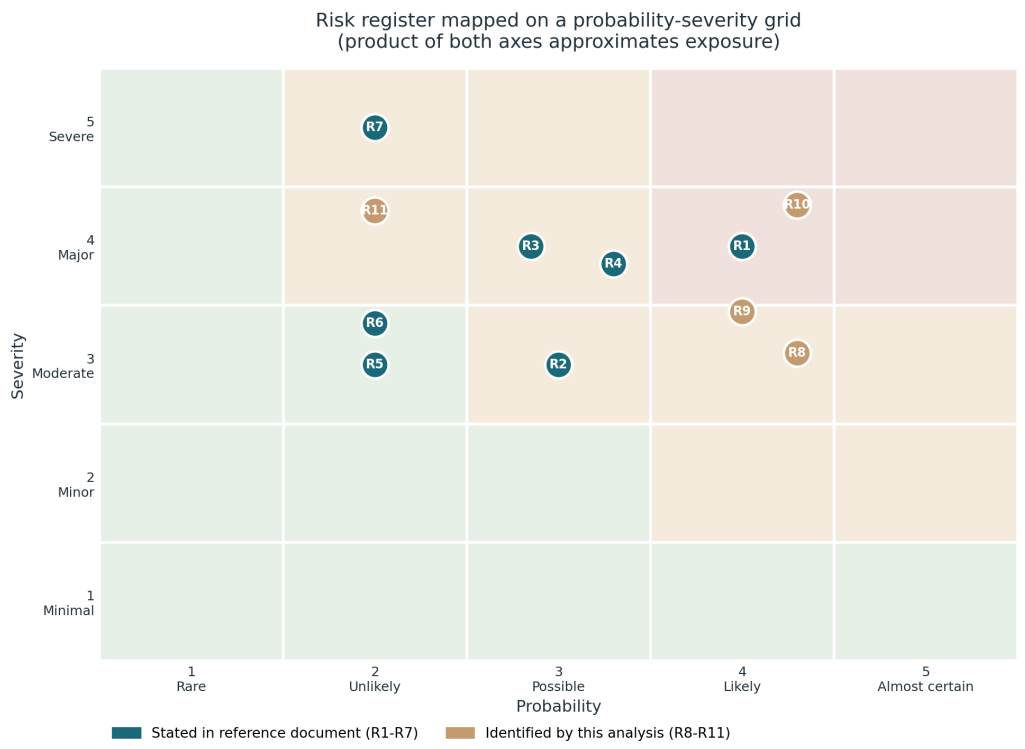
1. **Arithmetic:** nine phases sum to **18–26 weeks** against a 24-week window — zero buffer at maximum durations

2. **Composition:** the two most valuable contributions are the two most expensive — planner ablations need Phase 3 stable; the Arabic benchmark sits in months 5–6

3. **Concentration:** the question planner is a single-owner critical path (Member 1) — co-assign in month 3 (Action 8)

4. **Team:** sensible four-way split (LLM/interview · NLP/Arabic/data · backend/graph · frontend/QA), but execution-side risk is underweighted in the document.

Risk register — 11 risks; four identified by this analysis



RANKED BY EXPOSURE (P × S)

R10 **Benchmark produced too late** — **4.0 × 4.35**
start scenario production month 2; evaluate on the first 50 **(NEW)**

R1 **Scope explosion across 10 contributions** — **4.0 × 4.0**
freeze MVP and RQ set; enforce the descope ladder **(doc)**

R8 **Annotation quality, no IAA protocol** — **4.3 × 3.1**
written guidelines, double annotation on a sample, report kappa **(NEW)**

R9 **Evaluator recruitment fails in month 6** — **4.0 × 3.0**
book evaluators in month 2; pay or course credit **(NEW)**

R7 **Hallucinated requirements** — **2.0 × 5.0**
evidence-first design, validation rules, human approval **(doc)**

Legend: teal = stated in the reference document (R1–R7); tan = identified by this analysis (R8–R11). **Also watch R11 ethics/privacy (2.0 × 4.3):** request Track C permissions in week 1; Track A stands alone.

04 / 04

Actions & Recommendation

Twelve actions ordered by the deadline by which they stop being cheap — three horizons, one descope ladder, four conditions on the go decision.

12 actions

3 horizons

4 conditions

04

Twelve actions across three horizons

WEEKS 1–3 — Foundations

1. Freeze framing (A–D), RQ wording, descope ladder
2. Systematic-search protocol: queries, criteria, cutoff
3. Annotation guidelines + statistical analysis plan (power, seeds, IAA target)
4. Request Track C permissions; design Track A to stand alone
5. Graph-query spikes against the relational schema

MONTHS 2–3 — Momentum

6. Benchmark production cadence: 10 scenarios/week → 60–80 ready by month 4
7. Pilot utility ablation on the first 20 scenarios
8. Co-assign the planner; prompt versioning, logging, cost accounting
9. Book human evaluators (students and practitioners)

MONTHS 5–6 — Endgame

10. Hold the what-not-to-build list; integrations as demo videos only
12. Publication: benchmark + strategy comparison as a short paper

∞ ONGOING

11. Descope ladder as needed: N9 → N8 depth → N5 calibration → benchmark scale

The critical three: Actions 1, 3 and 6 — freeze the contract, define the statistics, start the benchmark now.

Recommendation: proceed — under four conditions

1. **Freeze the research contract** — RQs and the minimum viable contribution locked before Phase 3
2. **Move the benchmark left** — scenario production and annotation guidelines begin in month 2, not month 5
3. **Define the statistics first** — sample sizes, seeds and IAA thresholds before any data collection
4. **Adopt the descope ladder** — change-impact and requirement-debt features are sacrificed first, by rule

Never descoped

Question planner (N2) · Evidence model · **Arabic benchmark (N7)**

Descoped in order

N9 Requirements time travel → N8 Change-propagation depth → N5 Debt-score calibration → Benchmark scale

These conditions are not reservations about the idea — they are the discipline the document itself prescribes, applied to its own plan.

"No requirement gap is ever silently filled by a model — and no plan gap silently filled by optimism."

Conditional go · 4.0 / 5 — held to its own standard, RequirementPilot should be accepted with distinction and leave behind a citable benchmark and a replication-ready evaluation harness.

Month 2 — 60–80 scenarios + frozen RQs

Month 3 — pilot ablation results

Month 6 — defense + short-paper submission

Thank you — questions welcome